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Cover letter

Dear Editors of the *Journal of Applied Ecology*,

We are pleased to submit our manuscript entitled *"Wind energy development alters breeding habitat of the Bicknell's Thrush, a threatened range-restricted boreal songbird"* to the *Journal of Applied Ecology* for consideration.

The rapid expansion of renewable energy infrastructure is widely recognized as an important component of climate mitigation strategies, yet it also raises significant challenges for biodiversity conservation. In our study, we examine how wind energy development affects breeding habitat for the Bicknell's Thrush, a threatened boreal songbird classified as Vulnerable by the IUCN, and one of North America's most geographically restricted breeding songbirds.

Focusing on a region undergoing large-scale wind energy development since 2013 in Québec (Canada), located at the core of the species' breeding range, **our study provides the first quantitative assessment of how wind energy infrastructure alters breeding habitat for the Bicknell's Thrush**, with impacts disproportionately affecting the highest-priority habitat.

Importantly, **our study also demonstrates that significant opportunities existed within the study area to achieve the same wind energy capacity while avoiding impacts on the most sensitive habitat**, a result that holds even under scenarios favorable to wind energy development. Notably, two of the four wind farm sectors examined account for only a very small share of the total impact (despite having the same theoretical energy production). This contrast indicates that ecological impacts are not an inevitable consequence of wind energy production but depend critically on spatial siting decisions.

Finally, **our results highlight the importance of integrating high-resolution habitat data for species of conservation concern** early in the planning process, when siting decisions remain flexible and avoidable habitat loss can still be prevented.

We believe that these results are particularly relevant as wind energy projects and other energy infrastructures are rapidly expanding across Québec (and, more broadly, worldwide) and they provide spatially explicit information that can help decision-makers anticipate and better plan future developments with regard to the conservation of this at-risk species and, more generally, other species of conservation concern facing similar land-use conflicts. For

this reason, we believe that our manuscript will be of interest to the readership of the *Journal of Applied Ecology*.

This manuscript has not been published previously and is not under consideration for publication elsewhere. We declare that we have no competing interests.

Thank you for your consideration.

We look forward to your reply,

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